

2025MCM C题 经验分享



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上海交通大学

SHANGHAI JIAO TONG UNIVERSITY



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PART ONE

基本介绍





数据集



- [data_dictionary.csv](#) – database descriptions with examples
- [summerOly_athletes.csv](#) – all competitors with their sport, year, and result (medal type or none)
- [summerOly_medal_counts.csv](#) – complete country medal count tables for all summer Olympics from 1896 to 2024
- [summerOly_hosts.csv](#) – list of host country for all summer Olympics from 1896 to 2032
- [summerOly_programs.csv](#) – counts of number of events by sport/**discipline** and total for all summer Olympics from 1896 to 2032

Data, such as country designations, are recorded by the **International Olympic Committee (IOC)** (on their Olympics.com website) at the time of a given Olympics. Thus, designations may change in the data set. As with all data, there may be recording anomalies. Note, for example, in the athlete's data set in some cases for sports like tennis, table tennis, beach volleyball, the "Team" includes more detail than just the country. For example, Germany-1 would be the first of two beach volleyball teams from Germany in the 2000 Olympics. Decisions and assumptions about how to handle the data are an important part of the modeling process.



summerOly_athletes.csv

所有参加过比赛的运动员信息

Name, Sex, Team, NOC, Year, City, Sport, Event, Medal

summerOly_medal_counts.csv

每届比赛国家获得金银铜牌数和排名

Rank, NOC, Gold, Silver, Bronze, Total, Year

summerOly_hosts.csv

35届奥运会的年份和主办城市

Year, Host

summerOly_programs.csv

每个discipline下的逐年event数目

Sport, Discipline, Code, Sports Governing Body, 1896,



破题



- Develop a model for **medal counts for each country** (for Gold and total medals at a minimum). Include **estimates of the uncertainty/precision** of your model predictions and measures of how well model performs.
 - Based on your model, what are your projections for the medal table in the Los Angeles, **USA summer Olympics in 2028**? Include prediction intervals for all results. Which countries do you believe are most likely to improve? Which will do worse than in 2024?
 - Your model should include countries that have yet to earn medals; what is your projection for how many will earn their first medal in the next Olympics? What sort of odds do you give to this estimate?
 - Your model should also consider the events (number and types) at a given Olympics. Explore the relationship between the events and how many medals countries earn. What sports are most important for various countries? Why? How do the events chosen by the home country impact results?
- Athletes may compete for different countries, but it is not a simple matter for them to change due to citizenship requirements. Coaches, however, can easily move from one country to another as they do not need to be citizens to coach. There is, therefore, the possibility of a **“great coach” effect**. Two possible examples of this include Lang Ping^[2], who coached volleyball teams from both the U.S. and China to championships, and the sometimes-controversial gymnastics coach, Béla Károlyi^[3], who coached Romania and then the U.S. women’s teams with great success. Examine the data for evidence of changes that might be due to a “great coach” effect. **How much do you estimate such an effect contributes to medal counts**? Choose **three countries** and identify sports where they should consider investing in a “great” coach and estimate that impact.
- What other **original insight(s)** about Olympic medal counts does your model reveal? Explain how these insight(s) can inform country Olympic committees.

预测各个Discipline中有多少Event?
(SVR), Model I.



预测2028奖牌榜
(LSTM / XGBoost), Model II.

预测首次获得奖牌的国家
(Cox 比例风险模型), Model III.

对奖牌数做加权平均统计

Threshold Method + Isolation Forest
+ Linear Regression + Random Forest
(集成学习方法), Model IV.

水题：观察男女运动员奖牌比例



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PART TWO

求解方法





I、预测2028年奥运会各项赛事的数量



问题

- Your model should also consider the events (number and types) at a given Olympics. Explore the relationship between the events and how many medals countries earn. What sports are most important for various countries? Why? How do the events chosen by the home country impact results?

三个子问题:

- 项目数预测
- 分析项目重要性
- 主办国成绩影响

思考

- 如何依据往届比赛的项目数预测下一届比赛的项目数?
- 如何定义不同项目对国家的重要程度?
- 主办国的影响力如何体现?



- 1、利用机器学习做预测分析
- 2、进行项目重要性建模
- 3、分析历届主办国的成绩

模型

- 子问题一：项目数预测



- 采用回归模型求解

线性回归

弹性网络回归

梯度提升回归

线性回归

决策树回归

逐步回归

逻辑回归

支持向量回归

三阶高斯回归

岭回归

k临近回归

...

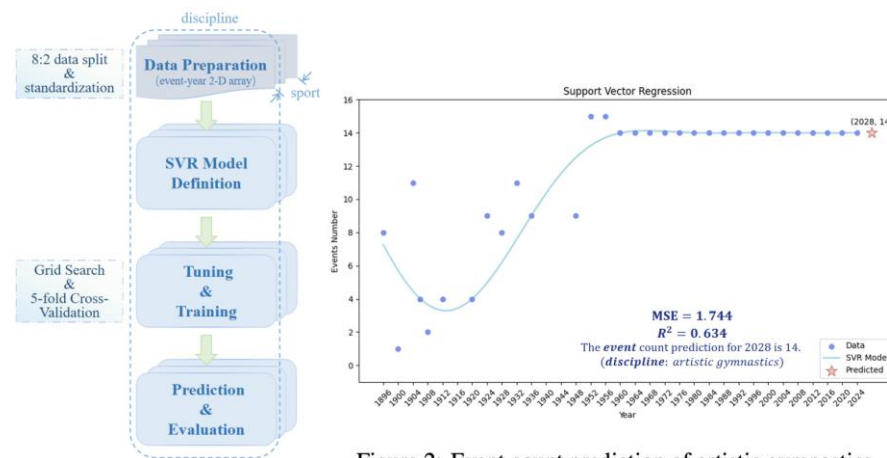


Figure 1: Flow chart of event count prediction

Figure 2: Event count prediction of artistic gymnastics



模型

- **子问题二：分析项目重要性** ➡ ➤ **进行项目重要性的建模**

Firstly, we quantified a country's performance in a specific sport at a certain Games as score S , which is the sum of gold, silver and bronze medal counts N_G , N_S and N_B assigned with different weights ω_G , ω_S and ω_B .

$$S = N_G \omega_G + N_S \omega_S + N_B \omega_B$$

For the sake of simplicity, we let

$$\omega_G = 3, \quad \omega_S = 2, \quad \omega_B = 1.$$



模型

- **子问题三：主办国成绩影响** ➡ ➤ **对历届主办国的成绩进行分析**

- 更熟悉场地
 - 减少了旅途奔波劳累
 - 更熟悉当地饮食
 - 观众的加油助威
- ➡
- 成绩提高

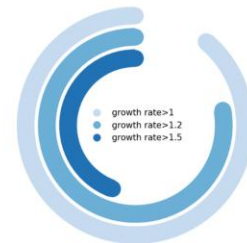
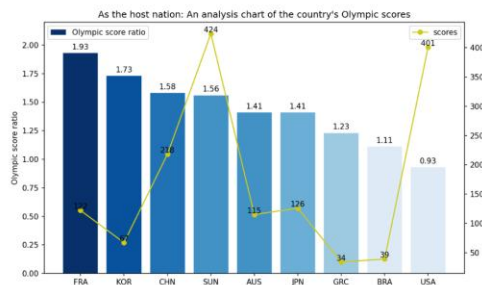


Figure 8: Analysis of the advantages of the host country. The light yellow score indicates the score when the country was the host country, and the score ratio indicates the score when the country is the host country divided by the score of the previous Olympics Game.



II、预测2028年奥运会各个国家获得奖牌数量

问题

- Based on your model, what are your projections for the medal table in the Los Angeles, USA summer Olympics in 2028? Include prediction intervals for all results. Which countries do you believe are most likely to improve? Which will do worse than in 2024?

两个子问题:

- 奖牌数预测
- 预测成绩显著变化的国家

思考

- 如何制定合理的奖牌预测方法?
- 如何找出最近实力提升/下滑显著的国家?

- 1、预测各个国家在每个项目的获奖概率
- 2、将获奖概率乘以项目数得到获奖数量
- 3、找到分析数据显著变化的算法

模型

- 子问题一：奖牌数预测

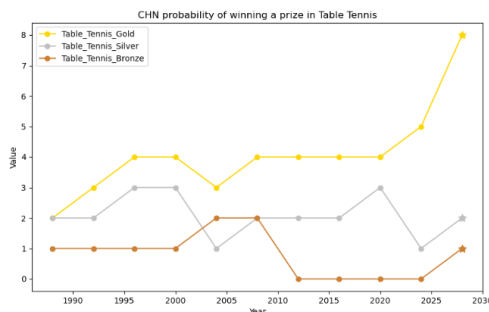


Figure 5: China's Table Tennis, Underfitting

- 采用LSTM+XGBoost的方法求解

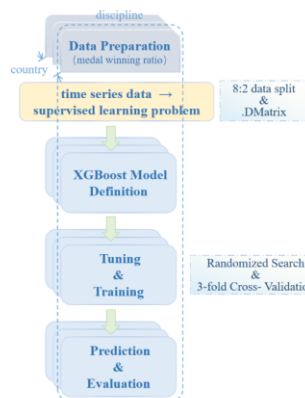


Figure 6: Flow chart of medal prediction process based on XGBoost

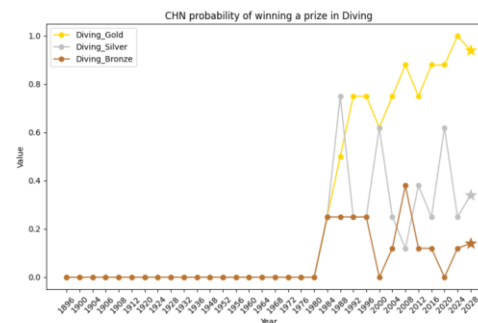


Figure 7: Probability prediction of Chinese diving gold and Silver MEDALS in 2028 Olympic Games

2028 USA summer Olympics				
Country				Total
USA	47	45	36	128
CHN	35	24	15	74
JPN	18	9	10	37
GBR	16	21	21	58
AUS	14	11	16	41
FRA	13	16	14	43
ITA	11	9	15	35
GER	11	11	12	34
NED	9	8	10	27
KOR	9	7	6	22

Figure 10: Top 10 of predicted 2028 medal table with host country effect



II、预测2028年奥运会各个国家获得奖牌数量



➤ 子问题二：预测成绩显著变化的国家 ➡ 查找论文找到分析数据变化情况的算法

➤ Mann-Kendall-based analysis → 抓取所有国家最近5届奥运会的成绩，利用算法找到奥运会成绩单调增和单调减的国家

模型

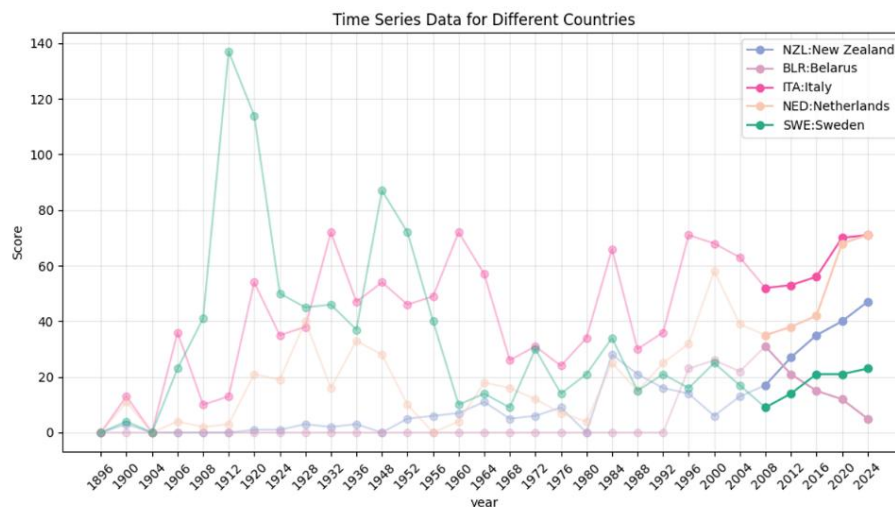
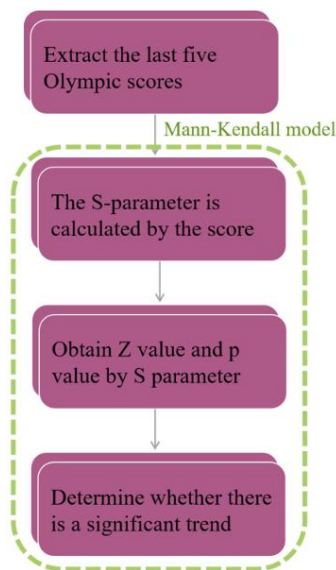


Figure 12: The scores of countries with obvious upward or downward trend in the past 5 Olympic Games



III、求解首次得牌国家



问题

- Your model should include countries that have yet to earn medals; what is your projection for how many will earn their first medal in the next Olympics? What sort of odds do you give to this estimate?

至少求解:

- 数量
- 概率

思考

- 整体趋势还是个体潜力，关注哪个更合理？
- 历史材料有限，该用哪些数据？
- 删失数据怎么办？



利用各个国家（未得牌）的数据，找到各个国家（未得牌）得牌的概率

模型

- 国家从奥运会开始到拿到第一个牌的过程 → 关注的过程
- 国家拿到第一个牌 → 发生目标事件
- 国家拿到第一个牌的概率 → 发生事件的概率
- 与国家实力相关的指标 → 影响因素

- 生存过程
- 风险事件
- 风险
- 预测变量

假设每个国家得牌的优势比例恒定（风险比恒定），获奖概率独立

通过非参数方式对基准风险函数进行建模，同时通过线性预测因子中的系数来估计预测变量的影响

Cox比例风险模型

Cox比例风险模型的目标在于估计风险函数，该函数表征在给定一组预测变量的条件下，特定时间点发生事件的瞬时风险。



III、求解首次得牌国家



利用Cox比例风险模型建模

$$P(t, X) = P_0(t) e^{\beta_a X_a(t) + \beta_p X_p(t) + \beta_e X_e(t)}$$

风险比

该国家在第t届奥运会
赢得首枚奖牌的概率

基准风险函数，从未参加过奥运赛
事的国家在第t届赢得首奖的概率

$X_a(t)$: → 人才数量、国家体育发展程度
第t届前累计参赛运动员总数 (不重复)

$X_p(t)$: → 国家奥运参与度和经验
第t届前累计参赛届数

$X_e(t)$: → 体育文化和教学资源多样性
第t届前累计参赛项目总数 (不重复)

实验仿真

模型拟合&预测

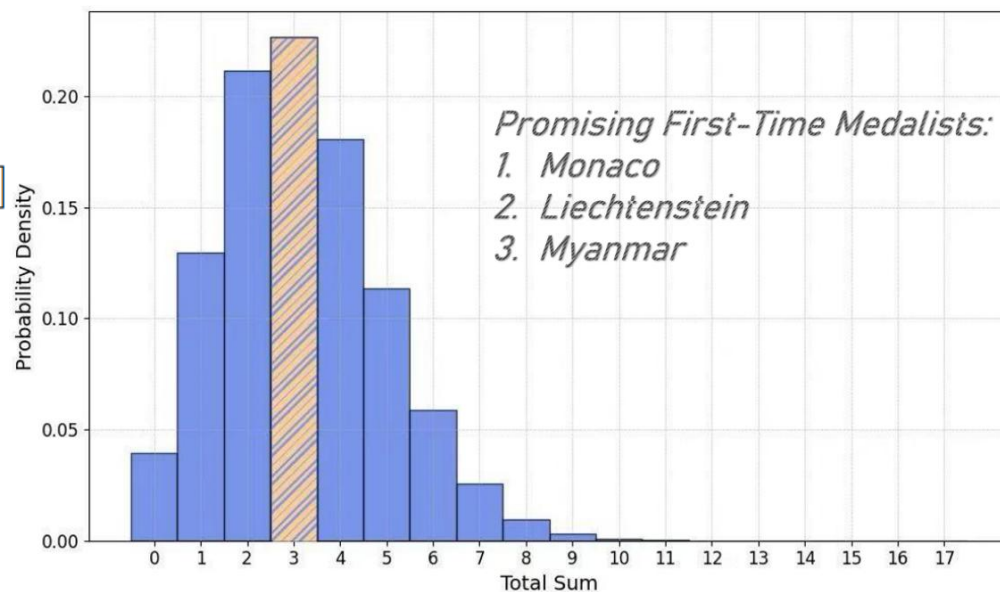
$$X = [X_a(t), X_p(t), X_e(t)]$$

- 逐届标记已获奖国家并进行生存分析
- 年份→持续时间变量，获奖→事件终点

蒙特卡洛仿真

$$Y(t) \sim (n(t), P(t))$$

- 在第t届奥运会进行n次独立重复试验，Y代表该国首次获奖的次数，则Y服从二项分布
- 基于预测的概率进行1亿次蒙特卡洛仿真



The distribution of possible number of 2028 first-time medalists.

NOC	Extrapolated Values	Exponential Values
MON	0.6995	0.0503
LIE	0.6434	0.0476
MYA	0.6297	0.0469
MLT	0.6215	0.0465
BOL	0.5682	0.0441
MLI	0.5519	0.0434
NEP	0.5464	0.0432
CHA	0.5452	0.0431
LBR	0.5343	0.0427
BIZ	0.5309	0.0425
CGO	0.5203	0.0421
AND	0.5163	0.0419
MAD	0.5066	0.0415
LES	0.5045	0.0414
GUI	0.4990	0.0412
BEN	0.4957	0.0410
NCA	0.4946	0.0410
SLE	0.4851	0.0406
CAY	0.4747	0.0402
CAF	0.4733	0.0401

Extrapolated and Exponential Values for top 20 first-time medalists in 2028.



VI、伟大教练效应建模



问题

Examine the data for **evidence** of changes that might be due to a “great coach” effect. **How much do you estimate such an effect contributes to medal counts?** Choose **three countries** and identify sports where they should consider investing in a “great” coach and estimate that impact.

三个子问题:

- 定义教练效应
- 分析影响
- 找三个国家推荐

思考

- 如何从给定的数据集中找到伟大教练效应?
- 如何分析伟大教练效应带来的影响?
- 如何说服特定的国家在某一体育方向中引入教练?



- 1、进行教练效应建模
- 2、进行教练效应的影响力建模
- 3、对教练能带来的提升进行建模

模型

- 子问题一：定义教练效应 ➡ 采用差分找异常值的方法

first-order difference and $S_{T\gamma}$ for the first-order ratio of scores between consecutive Olympic Games and compare the actual ΔS and γS with them.

$$GCE = \begin{cases} 1 & \text{if } \Delta S \geq S_T \parallel \gamma S \geq S_{T\gamma}, \\ 0 & \text{else.} \end{cases} \quad (20)$$

$$\Delta S = S_i - S_{i-1}$$

$$\gamma S = \frac{S_i}{S_{i-1}}$$

Points with a GCE value of 1 are defined as **differential points**.



VI、伟大教练效应建模



➤ 子问题三：分析影响



➤ 分析异常值占不同体育方向的比例和异常值占特定国家某一届奥运会总得分的比例

To measure the contribution of great coaches to a country's performance at the i^{th} Games, we defined a contribution factor C_i .

$$C_i = \frac{\sum_{j \in sports} \alpha_{ij} \Delta S_{ij}}{\sum_{j \in sports} S_{ij}} \quad (21)$$

If $\Delta S_{ij} > S_T$, $\alpha_{ij} = 1$; else $\alpha_{ij} = 0$.

To measure the influence of a coach to any country's performance in a certain sport s , we defined an influence factor I_s .

$$I_s = \frac{\sum_{k \in years} \sum_{countries \Delta S_{sk} > S_T} \alpha_{sk} \Delta S_{sk}}{\sum_{k \in years} \sum_{countries \Delta S_{sk} > S_T} S_{sk}} \quad (22)$$

If $\Delta S_{sk} > S_T$, $\alpha_{sk} = 1$; else $\alpha_{sk} = 0$. This factor mirror the relationship between coach and sports. It's different from sport to sport. Because the mobility of coaches is global, we assumed the I_s is the same for each country. We ranked I_s in table 5.

Table 4: NOC and their corresponding Contribution Factor

NOC	C_{2024}
VIE	1.0000
SGP	1.0000
CIV	1.0000
TTO	1.0000
ISL	1.0000
MAR	0.9318
CHI	0.9167
IRL	0.8635
POR	0.8590
SMR	0.8333
ANZ	0.8182
EST	0.8034
BOT	0.8000
BRN	0.8000
BAH	0.7894
MEX	0.7843
MGL	0.7500
NGR	0.7159
PHI	0.6875
URU	0.6778
ETH	0.6691
SLO	0.6666
HKG	0.6666
ESP	0.6611
DOM	0.6538
RSA	0.6427
ECU	0.6306
KEN	0.6266
PAK	0.6242
CRO	0.6306

Table 5: Sports and their corresponding Influence Factor

Sport	I_s
Cycling Mountain Bike	1.0000
3x3 Basketball	1.0000
Cycling BMX Racing	1.0000
Cycling BMX Freestyle	1.0000
Golf	0.8690
Surfing	0.8333
Triathlon	0.8161
Sport Climbing	0.8125
Trampoline	0.8077
Tug-Of-War	0.8068
Canoe Slalom	0.8065
Modern Pentathlon	0.8063
Cycling Road	0.8000
Art Competitions	0.7850
Beach Volleyball	0.7692
Handball	0.7590
Trampoline Gymnastics	0.7500
Football	0.7399
Water Polo	0.7358
Tennis	0.7324
Softball	0.7238
Volleyball	0.7211
Hockey	0.7124
Rhythmic Gymnastics	0.7063
Sailing	0.7058
Baseball	0.6989
Archery	0.6987
Artistic Swimming	0.6970
Shooting	0.6814
Taekwondo	0.6777

- 子问题二：找三个国家推荐 ➡ ➤ 采用线性回归和随机森林的方法分析引入教练会带来的影响

So we employed both Regression Analysis and Random Forest to fit a function, which we denoted as F , where the previous Game's score S_{i-1} is the independent variable and the following Game's score S_i is the dependent variable.

$$F(S_{i-1}) = S_i \quad (23)$$

F reflects the pattern of score improvement under Great Coach Effect. Once the sport and the previous score S_{i-1} are determined, the following year's score S_i can be predicted regardless of the country or year.

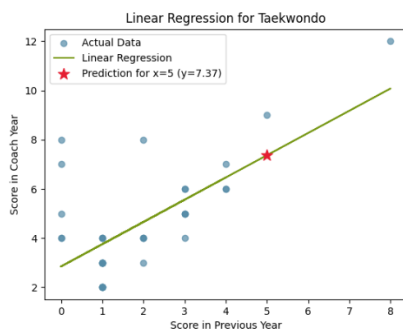


Figure 16: USA Taekwondo S prediction -Linear Regression

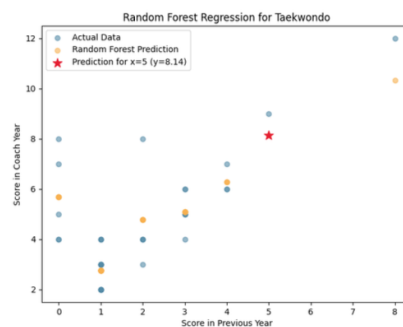


Figure 17: USA Taekwondo S prediction -Random Forest

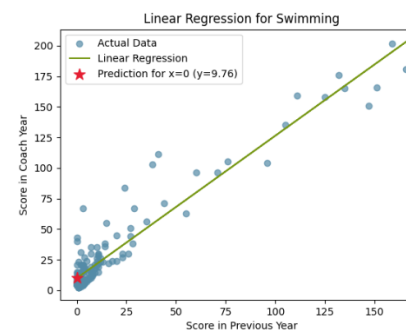


Figure 18: TPE Swimming S prediction -Linear Regression

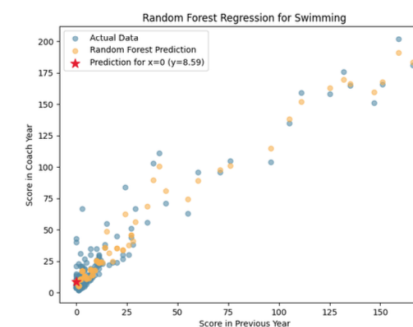


Figure 19: TPE Swimming S prediction -Random Forest



V、提出自己的见解



问题

- What other **original insight(s)** about Olympic medal counts does your model reveal? Explain how these insight(s) can inform country Olympic committees.

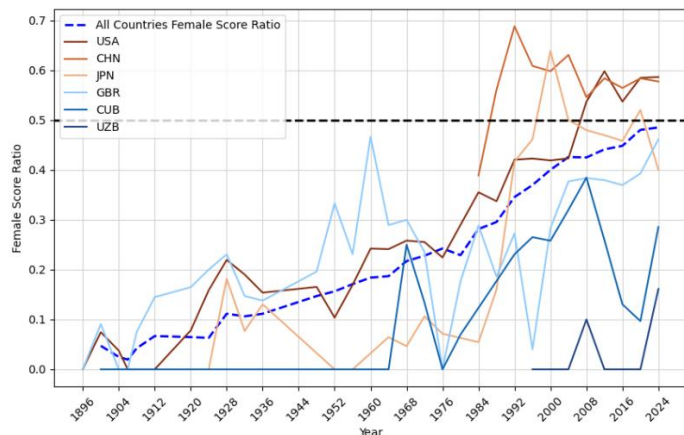
思考

- 可以分析获奖运动员中的男女占比?
- 历年获奖的男女占比的变化趋势是怎么样的?
- 是否奥运会强国的男女获奖比例更加均衡?



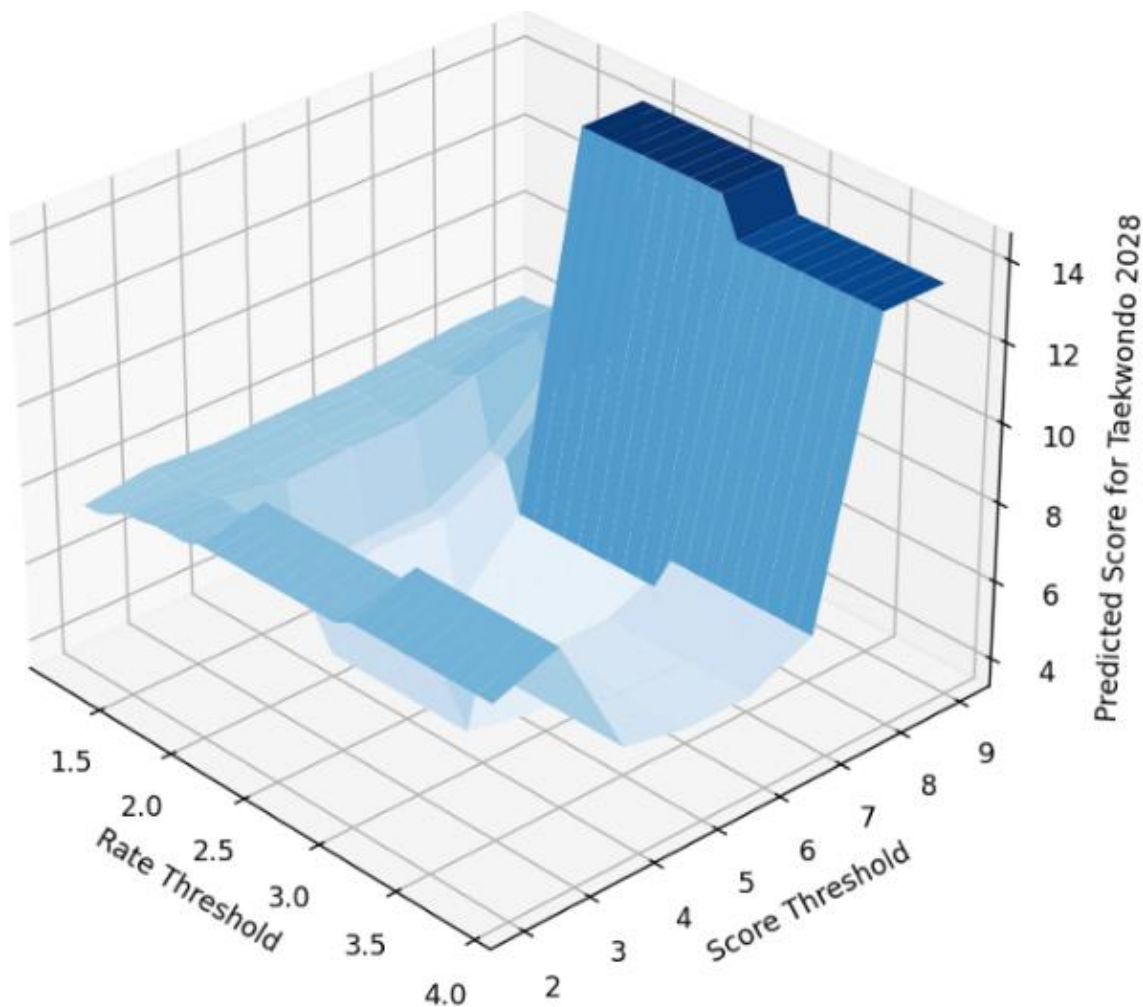
进行数据分析

模型



提出奥运会成绩没有那么好的国家可以尝试发展女性运动员的建议

Figure 20: Relationship between predicted score after hiring a coach and the thresholds



对雇佣教练后的预测分数 关于两个阈值进行敏感度分析

1. 模型对 Rate Threshold 较为不敏感,
2. 模型对 Score Threshold 敏感, 但是在我们的参数设置 (Score Threshold = 1.2) 处较为平缓,
3. 阈值升高意味着更强的教练被筛选出来, 预测分数随之升高符合预期,
4. 模型对阈值较为稳定的参数区间:

$$1.2 \leq \text{rate-threshold} \leq 4,$$

$$3 \leq \text{score-threshold} \leq 7.$$



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PART THREE

论文写作





写作过程



day1

背景&综述

- 分析题目
深层需求是什么?
需要攻克创新点在哪?
- 搜集资料为建模指明方向同时作为切题角度
数据源网页
文献综述
相似研究



day1-3

分析&建模

- 题目初步剖析后划分文章架构
分为几个板块来解决?
行文递进思路
- 确认模型可行后进行问题针对分析和建模具体阐释
需要做什么假设?
列出公式、示意图等



day2-4

实验&结果

- 编程初步得到结果时完成实验思路和方法部分的撰写
实验按什么步骤进行?
数据如何处理?
- 全面得到最终结果后进行结果展示和分析
验证结果合理性
重点结果可视化



day4-5

其余部分

- 并行进行
每个模型的overview、完善our work、assumptions、问题重述、notation...
- 最后进行
敏感性分析、额外工作、模型优缺点、附录、题目、摘要、AI报告...



写作过程



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背景&综述

理解到位，用语精炼，反复润色

分析&建模

起手生动，分析具体，结合公式

实验&结果

重点突出，图表丰富，现实意义

其他部分

持续更新，保持细致，保证耐力



写作注意事项



预先熟悉工具使用

- 掌握latex用法, 准备好论文模版
- 熟悉1-2种画图工具
- 熟悉多种数据库, 或使用文献管理软件, 确保能快速检索到论文

清楚论文架构细节

- 掌握论文模块组成
- 每部分大致需要的篇幅
- 注意细节, 比如角标、变量、图标、时态统一等...

学习往年获奖论文

- 了解破题思路
- 找到值得学习的优点并学以致用, 比如标题风格、措辞方式、可视化风格...

善用翻译工具

- AI润色、检查搭配用法是否得当, 确保表达精准、地道、没有冗余
- 对于特定词组、句式、短语等使用翻译工具, 以平时用的最顺手为宜

及时检查勘误

- 完成一个段落的撰写及时复查, 防止最后检查精力不足遗漏细节
- 写作过程中有不确定的地方标注出来, 及时与队友商讨

英语写作能力

- 多读论文熟悉学术英语写作方法
- 队里至少一人英语能力足够扎实



优势总结



1

问题分析全面且深入，有理有据

- **思路正确，解题完整**
按照解题思路自成体系，逻辑自治，不一定按照题目顺序
- **公式图表，解释充分**
每个公式、变量、下标和结果展示都要辅以精要解释
- **额外工作，对比优化**
同一问题多个模型对比，展现独到思考和优化思路

2

可视化丰富多样，图表信息充分、抽象与生动结合



3

语言表达到位，适当结合修辞为文章添彩留下印象



上海交通大学

SHANGHAI JIAO TONG UNIVERSITY

PART FOUR

参赛经历





赛前准备



组队情况

- ▶ **专业能力互补**
人工智能+微电子*2

- ▶ **角色经验互补**

编程手 → 建模手 → 写作手

深度协作、理解同步

备战情况

- ▶ **具备数模经验**
2024美赛C题F、2024国赛A题省一
- ▶ **参加数模培训营**
阅读获奖论文、学习经验谈、积累工具

做好准备、储备知识

比赛计划

- ▶ **确定场地和时间**
最好线下，便于交流，并且每天固定工作时间
- ▶ **确保进度合理**
提前对解题有大致规划
建议J人P人搭配

团结一致、确立目标

团队配合、齐心协力最重要！



赛中过程



建模手



编程手



写作手

选题读题，讨论大致怎么做，按什么顺序

想思路看文献

处理数据

看文献想思路

建模 I 编程 I

编程 II

开头和模型 I

建模 II 编程 II

编程 III

建模 III

建模 IV

编程 IV

模型 VI 模型 II

检查补充

摘要

其余部分

知识储备
+
学习能力
+
发散思维
+
善用辅助
+
实时交流
+
团结一心

赛中要点



线下讨论，思路同步



流水线工作，避免空闲



数据共享，可视化协作



遇到问题积极面对，灵活处理



赛后总结



ONE

团结

- 合理分配工作
- 严格规划时间
- 树立共同目标

TWO

耐力

- 合理分配精力
- 坚持到最后
- 保证充足休息

THREE

个性

- 学习往年论文
- 突出工作特色
- 给评审留印象

FOUR

严谨

- 模型的合理性
- 结果可解释性
- 研究的现实意义

谢谢大家
祝大家取得满意的成绩!

